

Predictive Analytics Report - Online Retail Dataset

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1. Introduction

This report presents a predictive analytics study conducted on the Online Retail dataset. The dataset contains transactional data from an online retail business, including invoices, products, quantities, prices, customers, and countries.

The analysis was performed using Python in a Jupyter Notebook environment. Followed a structured, systematic approach guided by the CRISP-DM methodology to ensure transparency and alignment with real business objectives. The workflow included data understanding, preparation, exploratory analysis, feature engineering, predictive modelling, and evaluation.

2. Dataset Description and Exploratory Analysis

2.1 Dataset Overview

The Online Retail dataset consists of transactional-level records with the following key variables:

- InvoiceNo: Unique invoice identifier (may repeat for multiple items in a single transaction)
- StockCode: Product identifier
- Description: Product description
- Quantity: Number of units purchased (negative values indicate returns)
- InvoiceDate: Date and time of transaction
- UnitPrice: Price per unit
- CustomerID: Unique customer identifier
- Country: Customer country

Each row represents a single product line item within an invoice, rather than the full transaction.

2.2 Data Quality Assessment

Before analysis, data quality checks were carried out:

Missing Values:

- Some CustomerID values were missing, which is common in retail datasets where purchases can be made anonymously.
- Transactions without CustomerID were excluded from customer-level predictive modelling, as identification is required to track repeat behavior.

Duplicate Records:

- Full-row comparison was used to detect exact duplicates across all attributes.
- 10,147 fully duplicated records were found, likely due to system re-entries or extraction artifacts. One copy of each transaction was retained, and duplicates were removed.
- Repeated invoice numbers alone were not considered duplicates, as they represent legitimate multiple product purchases in a single invoice.

Invalid Values:

- Transactions with negative quantities or unit prices (returns or cancellations) were excluded from predictive modelling, which focused on actual purchasing behavior.

2.3 Descriptive Statistics

Descriptive statistics were computed for Quantity, UnitPrice, and TotalSales (Quantity × UnitPrice).

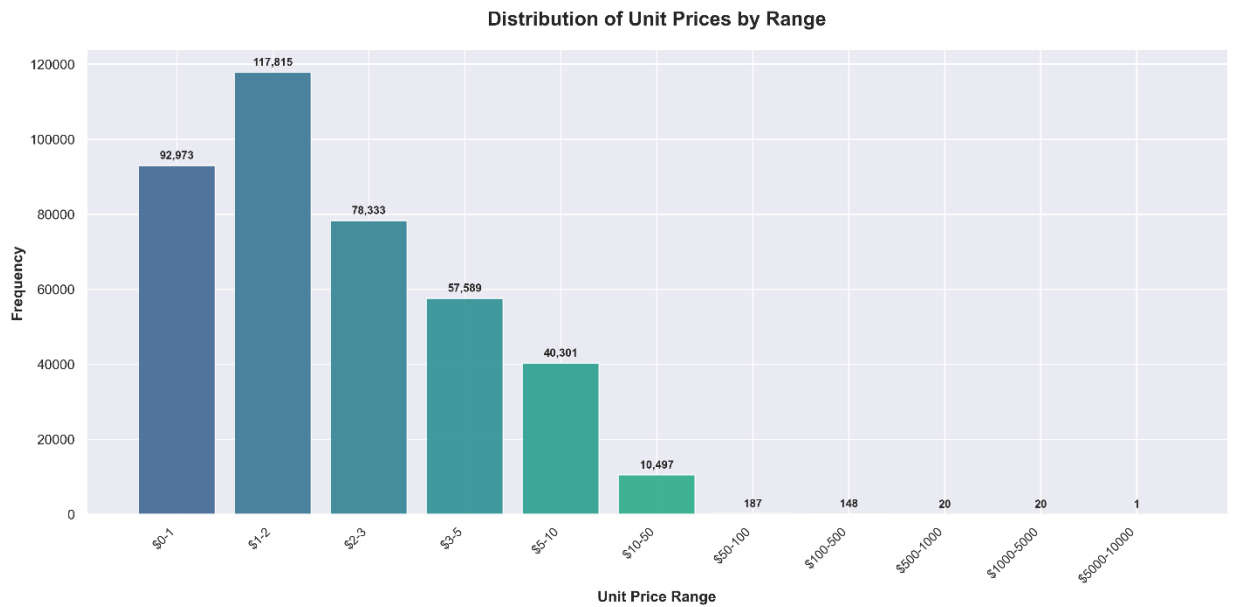
Key observations:

- Most transactions involve small purchase quantities.
- Unit prices are right-skewed, with a small number of high-priced items.
- Total sales values vary widely, reflecting different customer spending patterns.

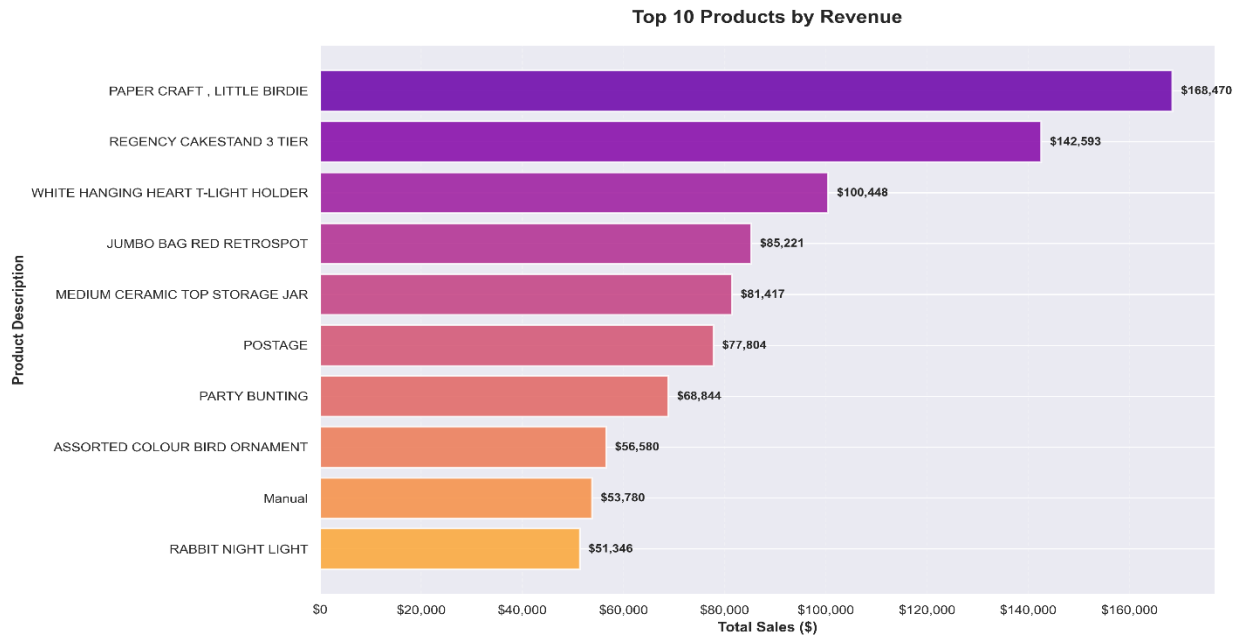
2.4 Exploratory Data Visualisation

Visualizations helped uncover trends and patterns in the data:

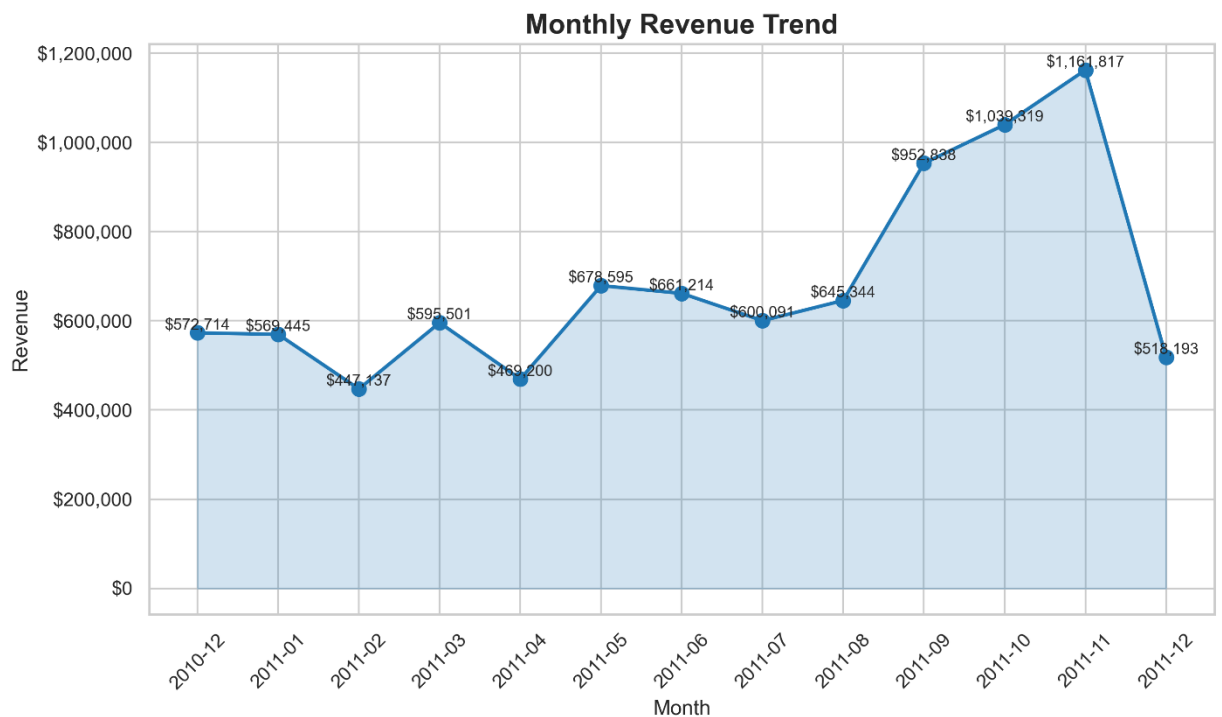
- Histogram of Unit Prices: Revealed a right-skewed distribution of product prices.



- Bar Chart of Top 10 Products by Revenue: Highlighted the most profitable products.



- Line Chart of Monthly Revenue: Showed seasonal trends and fluctuations in sales.



These charts provided valuable insights into product performance, customer purchase patterns, and temporal sales behavior.

3. Business Problem Definition

3.1 Business Context

Customer retention is critical for online retail businesses. Repeat customers tend to spend more and cost less to retain compared to acquiring new customers. Identifying customers likely to return allows the business to target marketing and loyalty strategies more effectively.

3.2 Business Problem Statement

The business problem addressed in this study is:

How can the retailer identify customers who are likely to make repeat purchases based on their historical transaction behavior?

3.3 Issue Tree Analysis

The main challenge—inaccurate sales forecasting—was analyzed using an issue tree:



Inaccurate Sales Forecasting

- Customer Behaviour
 - Repeat vs. one-time customers
 - Purchase frequency
- Product Demand
 - Seasonality
 - Product popularity
- Operational Factors
 - Stock availability
 - Pricing strategy

This analysis highlighted repeat purchasing behavior as a key driver of sales performance and a suitable focus for predictive modelling.

4. Predictive Task and Analytical Approach

4.1 Predictive Task Formulation

The predictive task is to identify whether a customer is likely to make repeat purchases based on their historical transaction behavior. This is framed as a binary classification problem:

- Target Variable: RepeatCustomer
 - 1: Customer has made more than one purchase
 - 0: Customer has made only one purchase
- Predictor Variables:
 - TotalSpend
 - AvgQuantity
 - NumPurchases

4.2 Modelling Approach

- The dataset was split into training (70%) and testing (30%) sets.
- Numerical features were standardised using a StandardScaler.
- A Logistic Regression model was trained to predict repeat customer status.

4.3 Model Evaluation

Model performance was evaluated using:

- Accuracy
- Confusion matrix
- Precision and recall

Key insights:

- Purchase frequency emerged as the strongest predictor of repeat behavior, followed by total spending.
- Customers who purchase more often and spend more are significantly more likely to return.
- While the model is not perfect, it provides actionable insights for marketing and customer relationship management, supporting data-driven retention strategies.

5. Conclusion

This study demonstrates the application of predictive analytics to real-world retail transaction data. By following a structured data mining methodology and leveraging Python-based analytical tools, meaningful insights into customer behavior were extracted.

The analysis shows that purchase frequency and spending behavior are strong indicators of repeat purchases. These insights can inform targeted marketing campaigns, loyalty programs, and overall customer retention strategies, helping the business make more data-driven decisions.

References

UCI Machine Learning Repository. *Online Retail Data Set*. University of California, Irvine, 2015, <https://archive.ics.uci.edu/dataset/352/online+retail>.

Wilson, E. (2021). *Predictive analytics for business forecasting & planning*. Graceway Publishing Company, Inc.